

Transcription - Factoid Sheet

Week 4 - Tuesday - Core Lecture 7 - current faculty deck + official Week 4 objectives

Master chain: promoter recognition → initiation → elongation → termination → eukaryotic cap/splice/poly(A) processing.

RNAP Basics

RNA synthesized **5' to 3'**; DNA template read antiparallel.

No primer required. RNAP initiates de novo; no 3' to 5' proofreading exonuclease.

Bacteria

Feature	Anchor
RNAP	Core: 2alpha + beta + beta' + omega; + sigma = holoenzyme
Sigma	Promoter specificity
Promoter	-35 and -10 / Pribnow box
Coupling	Transcription + translation coupled
Rho-independent termination	Hairpin + U-rich tract
Rho-dependent termination	Rho helicase binds rut; ATP-dependent catch/release

Eukaryotic RNA Polymerases

Pol	Products
I	45S pre-rRNA → 28S, 18S, 5.8S
II	pre-mRNA/mRNA, miRNA, many snRNAs
III	tRNA and 5S rRNA

Pol II Initiation

TBP helps assemble transcription complex. **TFIIH helicase** opens DNA; **TFIIH kinase** phosphorylates the Pol II CTD and promotes elongation.

Drug Targets

Agent	Target
Rifampin	Bacterial RNAP beta subunit
Alpha-amanitin	Euk RNAP II (III at higher dose)
Actinomycin D	Broad transcription inhibition

Transcription / Processing - Fast Recall

5' Cap

7-methylguanosine. Functions: nuclear export, protection from degradation, promotion of translation.

Splicing

Spliceosome = snRNPs + proteins. Key intron landmarks: **5' donor**, **branch site**, **3' acceptor**. Alternative splicing changes exon combinations.

Polyadenylation

AAUAAA signal → cleavage ~10-30 nt downstream → poly(A) polymerase adds ~80-250 A residues. Tail supports termination, export, stability, translation.

Alternative Polyadenylation

Different 3' cleavage sites change 3' UTR length. Shorter 3' UTR can remove miRNA sites → more translation; longer 3' UTR can retain more regulatory sites.

Disease Anchors

Condition	RNA-processing link
Autosomal-dominant retinitis pigmentosa	Several disease genes encode pre-mRNA splicing factors (PRPF family)
SLE	Autoantibodies can target snRNP/Sm spliceosomal components

Prok vs Euk

Bacteria	Eukaryotes
1 RNAP; sigma; -10/-35	Pol I/II/III; general TFs; enhancers
Coupled transcription/translation	Spatially separated
Often polycistronic	Usually monocistronic
No nuclear cap/splice program	5' cap + splicing + poly(A)

30-Second Review

No primer. Sigma finds bacterial promoters. Rifampin = bacterial beta. Pol I rRNA; II mRNA; III tRNA/5S. Alpha-amanitin = Pol II. Euk mRNA = cap, splice, poly(A), export.

Assigned ScholarRx

DNA Transcription & Translation: <https://exchange.scholarrx.com/brick/dna-transcription-and-translation>